Cross-Border Competition and Tax Pass-Through: Evidence from Germany's 2022 Fuel Tax Cut

Mats Petter Kahl

Universidad Carlos III de Madrid, Department of Economics, C/ Madrid 126, 28903 Getafe (Madrid), Spain

Corresponding author: mkahl@eco.uc3m.es · ORCID 0000-0003-3387-6386

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Abstract

How does a national tax cut pass through at an international border when one side is treated, and the other is not? I study Germany's 2022 fuel-tax cut using station-level price data and exact driving distances to French competitors. German diesel stations within 10 km of a French gas station cut prices by 0.96 cents per liter more than stations further than 50 km inland. That corresponds to a pass-through premium of about 6 percentage points. The premium appears on the first day of the cut, is largest within a few kilometers of the border, and disappears after the tax returns. The premium arises where the tax cut closed the cross-border price gap, that is, for diesel, and it doesn't appear where the gap stayed wide, that is, for gasoline. A model with treated domestic and untreated foreign rivals shows that, at a given intensity of cross-border substitution, feedback through foreign prices explains at most a small part of the premium. The evidence is consistent with a change in that intensity: cross-border competition that becomes active when the price gap is small. The incidence of a national tax cut therefore varies within a country with local competitive conditions, here, at the border.

Keywords: tax pass-through; tax incidence; fuel prices; border markets; retail competition

JEL classification: H22; H73; L13; R32

1 Introduction

How does a national tax cut reach consumers at an international border? A unilateral tax change alters both the local price level and the relative price across the frontier. Motorists can buy on either side, so a border station competes with rivals whose costs fell and with rivals whose costs did not. Pass-through may therefore differ between border and inland stations, and prices may move even where the tax is unchanged.

Germany's 2022 fuel discount provides a useful setting. From 1 June through 31 August, the federal government reduced the energy-tax rate by 29.55 cents per liter for gasoline and 14.04 cents for diesel. Including the associated value-added-tax saving, full pass-through implied price reductions of 35.16 cents for gasoline and 16.71 cents for diesel (Bundesministerium der Finanzen, 2022). France maintained a separate 18-cent tax-inclusive rebate throughout the German policy window (République française, 2022a,b). The German reform therefore changed the bilateral price gap without a contemporaneous change in the French statutory rebate.

I combine the reform with granular geography and timing. The German data cover reported price changes in Baden-Württemberg, Rheinland-Pfalz, and Saarland; the French data come from the official station-price registry. Stations are matched to driving distance from the nearest competitor across the French–German border. The principal specification compares country-by-distance groups before and during the reform while absorbing permanent station differences and common daily shocks. Because each record identifies when a posted price became valid, I reconstruct duration-weighted daily price paths rather than average irregularly timed quotes.

The central result is a within-Germany contrast. Diesel prices at German stations within 10 km of a French station fall by 14.18 cents per liter relative to inland France, while prices more than 50 km from a French competitor fall by 13.22 cents. The 0.96-cent difference has a 95% confidence interval of 0.43–1.49 cents and equals 5.8 percentage points of the tax-inclusive cut. The differential appears when the reform begins, remains negative, and is stable when the near-border threshold is set to 8 or 12 km. It is not a generic border effect: for E5 and E10 gasoline, pass-through is high but the nearest-ring estimates are 1.8–1.9 percentage points below the inland estimates and are not statistically distinguishable from them.

The timing points to the reform. In a daily event study, the difference between near-border and inland prices falls by 1.35 cents on the first day of the cut relative to the last day before it, and by 1.95 cents relative to the average of the preceding three weeks. Weekly pre-reform coefficients are individually small, although they reject jointly; bounds that allow post-reform trend violations as large as the largest pre-reform violation (Rambachan and Roth, 2023) still exclude zero for the first policy week. When the tax returned on 1 September, the premium disappeared within about three weeks. It is largest within a few kilometers of the border and survives controls for brand, motorway location, the number of nearby rivals, and urban–rural type, as well as county-by-date fixed effects, which compare border stations with other stations in the same county on the same day. The premium is precisely estimated with station-clustered errors and remains significant at the 10% level when errors are clustered by county.

The pattern across fuels and over time fits a simple account of cross-border competition. A

foreign station constrains a border station only when motorists are close to indifferent between them, that is, when the cross-border price gap is small. Before the cut, German diesel cost about 12.5 cents more than French diesel; the cut nearly closed that gap, and the premium appeared. For gasoline, the cut turned a German disadvantage of 19–20 cents into an advantage of 11–13 cents; the gap reversed but stayed wide, cross-border competition stayed inactive, and no premium appeared. When the German tax returned and France raised its rebate on 1 September, German diesel again became much dearer than French diesel, and the premium vanished. The evidence points to competition across the border that was active when the gap was small and inactive when it was large.

To interpret the premium, I develop a model with a treated border station, a treated domestic rival, and an untreated foreign rival. The untreated rival lowers its price in response to the German cut, and this response feeds back into German pass-through, but only at second order. The border premium is therefore a net object: foreign feedback raises it, while weaker exposure to treated domestic rivals lowers it. Using the French price response, the model bounds the foreign-feedback part of the premium at 0.3–0.5 percentage points, and at most 1.7 points at the upper end of the confidence interval, against a premium of 5.8 points. At a given intensity of cross-border substitution, feedback through foreign prices therefore explains at most a small part of the premium. The evidence is consistent with a change in that intensity: competition across the border that became active when the cut closed the diesel price gap and inactive when the gap reopened. The model holds that intensity fixed, so it frames this account rather than deriving it.

French diesel prices within 30 km of Germany also fell relative to inland France, the direction the model predicts for an untreated rival. French border prices were already drifting down relative to inland France before the reform, however, so I do not interpret the size of the French response and use it only as an upper limit in the bound.

The article contributes to work on tax incidence under imperfect competition (Anderson et al., 2001; Weyl and Fabinger, 2013; Ritz, 2024), on the pass-through of own and rival cost shocks (Muehlegger and Sweeney, 2022), and on incidence near jurisdictional borders. Tax differences shape where consumers buy and where retail activity locates: Manuszak and Moul (2009) show that tax differences between Chicago, its suburbs, and Indiana shift gasoline station capacity toward lower-tax jurisdictions, and Agrawal (2015) documents a gradient of local tax rates with driving time to a higher-tax state. Doyle Jr. and Samphantharak (2008) find that pass-through of a US gasoline-tax moratorium was smaller near state borders and that its reinstatement raised prices up to an hour's drive into neighboring states; Harding et al. (2012) find that cigarette-tax pass-through rises with distance from a lower-tax border; and Bajo-Buenestado and Borrella-Mas (2019) document a price response outside the taxing jurisdiction after a Portuguese fuel-tax increase. Kahl (2024a) shows that large tax and price differentials at the German–Polish border lead to inactive price competition and one-directional border crossing, as well as systematic differences in retail infrastructure in border regions, and Kahl (2024b) documents differences in pass-through of the same German tax cut over space and time. Relative to this work, the article studies a national tax cut at an international border with station-level prices observed on both sides and driving distances between stations. It separates treated from untreated rivals in a pass-through model, bounds the foreign-feedback channel, and relates the border premium to the cross-border price gap across fuels and over the policy cycle.

French card transactions show that fuel demand at this border responds strongly to relative prices (Adam et al., 2024).

Section 2 describes the policy setting. Section 3 develops the model, Sections 4 and 5 present the data and design, Section 6 reports the results, and Section 7 evaluates identification and robustness. Section 8 concludes.

2 Institutional setting and related evidence

2.1 The German tax cut and the French rebate

Germany's reform temporarily lowered statutory energy-tax rates to the European minimum. Because value-added tax is levied on the tax-inclusive base, the consumer-price cost wedge fell by 16.71 cents per liter for diesel and 35.16 cents for E5 and E10 gasoline. The original rates returned on 1 September 2022 (Bundesministerium der Finanzen, 2022). Retail prices were not administratively regulated.

France operated a separate rebate. In metropolitan France it equaled 15 cents before tax, or 18 cents including tax, from 1 April through 31 August 2022 (République française, 2022a,b). A decree issued on 22 August raised the tax-inclusive rebate to 30 cents on 1 September, the same day the German cut expired (République française, 2022c). TotalEnergies added its own rebate of 20 cents per liter at its French stations from the same day (Motor1, 2022), and strikes at French refineries disrupted supply from late September (Agence France-Presse, 2022). The end of the German cut was therefore a combined event. The final treatment weeks can reflect expectations about it, and Section 7.2 uses the expiry as a reversal test that accounts for the French changes.

Other measures and shocks overlapped with the reform. The French rebate was initially scheduled to end on 31 July and was extended to 31 August only by the decree of 23 July (République française, 2022b). Luxembourg lowered fuel prices by 7.5 cents per liter from 13 April 2022, initially until 31 July and eventually through 31 August (Gouvernement du Grand-Duché de Luxembourg, 2022; L'essentiel, 2022). Germany's relief package also offered the 9-euro public-transport ticket in June, July, and August 2022 (Krämer et al., 2022). Germany temporarily reintroduced controls at its internal borders from 13 June to 3 July 2022 for the G7 summit (European Commission, 2026). In August, low water on the Rhine hampered fuel transport and contributed to higher prices in southern Germany (Bundeskartellamt, 2022). Finally, the energy tax is levied when fuel leaves the refinery or tax warehouse, not at the pump (Bundesregierung, 2022), so stations entered June with stocks taxed at the old rate.

2.2 Cross-border choice

Let a motorist's generalized cost of buying abroad equal the posted foreign price plus travel and time costs. Whether the foreign station constrains a domestic seller therefore depends on distance and on the bilateral price gap, not distance alone (Kanbur and Keen, 1993; Chandra et al., 2014; Jansen and Jonker, 2018). When the gap is large, border crossing runs in one direction and price

competition across the border becomes inactive (Kahl, 2024a). This extensive margin is absent from the covered-market Hotelling benchmark, in which distance changes markups but need not change cost pass-through (Hotelling, 1929). The model below takes the intensity of cross-border substitution as given; how it varies with distance and the price gap is discussed but not modeled.

2.3 Evidence on the 2022 fuel discount

Published estimates generally indicate substantial average pass-through, with variation by fuel, period, and design (Dovern et al., 2023; Drolsbach et al., 2023; Bernhardt et al., 2023; Schmerer and Hansen, 2023; Seiler and Stöckmann, 2023; Kahl, 2024b). Freitas and Syga (2022), Frondel et al. (2026), and Kahl (2024b) document pronounced time and spatial heterogeneity. von Waldow and Link (2026) relate pass-through to horizontal and vertical market structure, while Genakos and Pagliero (2022) show in isolated fuel markets that pass-through rises with the number of competitors. The present design asks a different question: how does incidence change when some nearby rivals receive the tax cut and others, across the border, do not?

3 A model of border tax incidence

3.1 Treated and untreated rivals

Consider a German border station G, a nearby German domestic rival D, and a French rival F. For each station $j \in \{G, D, F\}$, p_j is the consumer price and q_j is the quantity sold. If v is the value-added-tax rate, c_j is resource cost, and t_j is the excise (fuel) tax, define tax-inclusive marginal cost as $m_j = (1 + v)(c_j + t_j)$. Multiplying profit by $1 + v$ does not change the price choice, so station j maximizes $(p_j - m_j)q_j$. Let $\Delta t > 0$ denote Germany's excise-tax reduction and $\Delta\tau = (1 + v)\Delta t$ its tax-inclusive value. The reform changes both German stations' marginal costs by $-\Delta\tau$; French marginal cost is unchanged.

Demand is linear:

$$\begin{aligned} q_G &= a_G - bp_G + b\delta_B p_D + b\rho p_F, \\ q_D &= a_D - bp_D + b\delta_B p_G, \\ q_F &= a_F - bp_F + b\rho p_G, \end{aligned} \tag{1}$$

Here a_j is a local demand intercept and $b > 0$ is the common own-price slope. The normalized slope $\delta_B \in [0, 1)$ measures substitution between the two treated German stations, while $\rho \in [0, 1)$ is the local cross-price slope between the German border station and the untreated French station, which I call the intensity of cross-border substitution. The common ρ is consistent with the symmetry of an integrable quasi-linear demand system: the same marginal motorists move between G and F in either direction. Integrability with strictly concave utility requires $\delta_B^2 + \rho^2 < 1$, which also guarantees the equilibrium condition in Appendix A. I maintain this restriction for the benchmark;

the feedback calculation below implies a small ρ, but the article does not estimate δ_B or δ_I (the inland counterpart of δ_B, defined in Section 3.2). The specification is a linear differentiated-products Bertrand model (Singh and Vives, 1984) applied to the incidence problem (Anderson et al., 2001); it differs from a two-firm model by distinguishing a rival that shares the German cost shock from one that does not. Because demand curvature affects pass-through (Weyl and Fabinger, 2013; Miklós-Thal and Shaffer, 2021), the comparative-static results below are specific to the linear-demand model and need not generalize to nonlinear demand.

The cross-price slope is not the share of motorists who buy abroad. Let x denote driving distance to the foreign alternative and $z = p_G^e - p_F^e$ the expected German-minus-French price gap. In a smooth random-utility or travel-cost model, the foreign purchase share is governed by a distribution function, whereas $\rho(x, z)$ is proportional to the density of motorists at the switching margin. It can therefore vary with distance and the price gap and can be hump-shaped in z. French card transactions confirm that the switching margin responds strongly to relative fuel prices at this border (Adam et al., 2024). The model holds ρ fixed; Section 3.3 discusses what changes when it moves with the price gap. Appendix A gives the stability condition.

3.2 Pass-through and the border premium

The first-order conditions, divided by b, are

$$\begin{aligned} 2p_G - \delta_B p_D - \rho p_F &= a_G/b + m_G, \\ 2p_D - \delta_B p_G &= a_D/b + m_D, \\ 2p_F - \rho p_G &= a_F/b + m_F. \end{aligned} \tag{2}$$

Let τ index the common tax-inclusive component of the two German stations' marginal costs, and let $d\tau$ be an increase in that component, holding the local slopes fixed. Solving the system gives

$$\Pi_B \equiv \frac{dp_G}{d\tau} = \frac{2 + \delta_B}{4 - \delta_B^2 - \rho^2}, \qquad \frac{dp_F}{d\tau} = \frac{\rho}{2}\Pi_B. \tag{3}$$

The first expression, Π_B, is the pass-through rate at the German border station: the change in its price per unit change in the German stations' tax-inclusive cost. The second is the price response of the untreated French station to the same change. Prices are strategic complements, so an exposed French station lowers its price even though its own cost is unchanged (Bulow et al., 1985). This is the model's direct cross-border spillover and parallels the price transmission outside the taxing jurisdiction documented by Bajo-Buenestado and Borrella-Mas (2019). The French response is first order in ρ; the feedback from F to German pass-through enters through ρ^2. Price data alone cannot separate the underlying demand diversion from the French best response.

For comparison, let an inland German station face a treated domestic rival with substitution weight δ_I and no foreign rival. Its common-shock pass-through is

$$\Pi_I = \frac{1}{2 - \delta_I}. \tag{4}$$

Unlike the two-firm model, this benchmark permits pass-through between one-half and one as treated domestic competition strengthens. Comparing (3) and (4), border pass-through exceeds inland pass-through if and only if

$$\rho^2 > (2 + \delta_B)(\delta_I - \delta_B). \tag{5}$$

Equation (5) isolates two opposing forces. If the foreign station is additional competition and treated domestic exposure is unchanged ($\delta_B = \delta_I$), cross-border feedback raises pass-through. If access to the foreign station replaces treated domestic competition ($\delta_B < \delta_I$), the border station faces fewer rivals whose costs fall; this lowers pass-through unless cross-border feedback is strong enough. Evidence that own- and rival-cost shocks pass through differently supports this distinction (Muehlegger and Sweeney, 2022).

Proposition 1 (Border incidence)**.** *With linear demand and fixed slopes, at a stable equilibrium a German tax cut lowers p_G and weakly lowers p_F, with a strict French decline when $\rho > 0$. Holding domestic competition δ_B fixed, cross-border substitution raises German pass-through because $\partial \Pi_B / \partial \rho > 0$ for $\rho > 0$. The border premium is positive if and only if* (5) *holds. If $\delta_B = \delta_I$, any $\rho > 0$ generates a positive but second-order premium; if $\rho = 0$, the sign of the premium is the sign of $\delta_B - \delta_I$.*

The proposition separates two empirically distinct margins. A French response is the first-order implication of an untreated rival's best response. A German border premium is a net object: it combines that foreign feedback with differences in treated domestic competition and therefore does not, by itself, identify the strength of cross-border substitution. Previous tax studies find lower pass-through near a lower-tax border (Doyle Jr. and Samphantharak, 2008; Harding et al., 2012); in the model, such a pattern arises when weaker treated-rival exposure dominates foreign feedback.

How large can foreign feedback be? The model bounds the part of the German premium that can come from feedback through French prices without using pass-through levels, which are not identified against inland France (Section 7). Suppose treated domestic competition is the same at the border and inland ($\delta_B = \delta_I = \delta$), so that any premium is due to foreign feedback. Equations (3) and (4) then give a premium of $\rho^2 \Pi_B / (4 - \delta^2)$. Writing f for the French price response, equation (3) gives $\rho = 2f / (\Pi_B \Delta\tau)$, so the feedback premium equals $4(f/\Delta\tau)^2 / [(4 - \delta^2)\Pi_B]$, which is approximately $4(f/\Delta\tau)^2/(2 + \delta)$ when ρ is small. With the 0–10 km French estimate $f = 0.81$ cents and $\Delta\tau = 16.71$ cents, the feedback premium lies between 0.31 and 0.47 percentage points for $\delta \in [0, 1)$; at the upper end of the 95% confidence interval for f (1.55 cents) it is at most 1.7 points. At a given intensity of cross-border substitution, feedback through French prices therefore explains at most a small part of the 5.8-point premium. If each border station faces several French rivals, the feedback term scales with the number of rivals on each side, but explaining the full premium would require a multiplier of 12 to 19. Within the linear model at fixed slopes, the remainder must reflect differences in treated domestic competition between border and inland stations. Outside it,

the remainder may reflect a change in the intensity of cross-border substitution as the cut moves the price gap (Section 3.3), demand curvature at the border, or differences in station composition, which Section 7 examines. The bound treats French stations more than 50 km from Germany, against which f is measured, as unaffected by the German cut. Because French border prices were already drifting down relative to inland France before the reform (Section 7.1), the German-caused part of f is likely smaller than its estimate, which makes the bound conservative.

3.3 Empirical predictions and scope

The model organizes the evidence into three comparisons. First, an untreated French price response in the direction of the German cost shock is the sharpest sign prediction. Second, the German near-minus-inland coefficient measures the net of foreign feedback and treated domestic competition; theory does not impose its sign. Third, effects can differ across fuels and policy phases because the intensity of cross-border substitution, $\rho(x, z)$, is evaluated at different price gaps. When the gap is small, many motorists are close to indifferent between a German and a French station, and cross-border competition is active; when it is large, few are, and competition is inactive. The linear model does not sign this state dependence, so it serves as an interpretation of the contrasts across fuels (Section 6.2) and over the policy cycle (Section 7.2).

Equations (3)–(5) are comparative statics at fixed slopes; with linear demand they are exact for a discrete change in costs as long as the slopes stay fixed. If the reform moves the price gap and with it ρ, the total price response adds a term $(\partial p_G/\partial \rho)(d\rho/d\tau)$ whose size depends on markups and on the curvature of demand at the switching margin, and which the linear model does not sign (Weyl and Fabinger, 2013; Ritz, 2024). The model is therefore used for the sign of the French response, to bound the feedback channel, and to frame the active-competition interpretation, not to fit pass-through levels or every ring coefficient. It abstracts from multiproduct ownership, wholesale contracts, dynamic search, and intertemporal refueling.

4 Data and price construction

4.1 Sources and sample

German price changes come from the Market Transparency Unit for Fuels and are distributed as historical files by Tankerkönig under a Creative Commons license (Tankerkönig, 2022). The reporting system covers E5, E10, and diesel prices at stations subject to the German reporting obligation. French prices come from the official *prix-carburants.gouv.fr* open-data portal (Ministère de l'Économie, des Finances et de la Souveraineté industrielle, 2022). The analysis covers 14 April–31 August 2022: 48 pre-policy days and the full 92-day treatment period. For the reversal test in Section 7.2, the German panel is extended with the same construction to 26 September 2022, the day before the French refinery strikes began.

The spatial sample contains the 2,880 German stations in Baden-Württemberg, Rheinland-Pfalz, and Saarland and the 769 French stations in the ten départements closest to Germany: Moselle, Bas-

Rhin, Haut-Rhin, Meurthe-et-Moselle, Vosges, Territoire de Belfort, Meuse, Haute-Marne, Haute-Saône, and Doubs. French stations more than 50 km from Germany within these départements form the reference group. Each German station is linked to the nearest French station by driving distance; French stations are analogously linked to Germany. Driving distances and times come from the Open Source Routing Machine (Luxen and Vetter, 2011), queried from R (Giraud, 2022) with the car profile on a merged OpenStreetMap road network for France, Germany, Luxembourg, and Switzerland as of November 2025 (OpenStreetMap contributors, 2025). Figure 1 makes the resulting market geography visible. Stations are dense on both sides of the Rhine, but feasible cross-border alternatives are unevenly distributed along the frontier. The design therefore uses driving distance to an actual foreign station rather than distance to the political border. The map also displays Luxembourg and Switzerland because routes and local market geography are not confined to the bilateral border. Prices from those countries are not used in estimation.

The diesel estimation sample contains 400,250 German and 107,541 French station-days, or 507,791 observations. The E5 and E10 samples contain 437,324 and 467,452 observations. A further 1,078 German diesel station-days have no valid routing distance: they enter the broad descriptive series but not the regressions. Tables C.1 and C.2 report descriptive statistics for E5 and E10.

4.2 Duration-weighted daily prices

German retail prices can change repeatedly within a day (Haucap et al., 2017; Kahl and Wein, 2025); French stations adjust less frequently on average (Gautier and Le Saout, 2015). A simple mean of posted quotes would therefore overweight intervals with frequent changes. I reconstruct a piecewise-constant price path for every station, fuel, and day. The last valid quote observed by 06:00 seeds the daily path; subsequent quotes update it. The daily outcome is the duration-weighted mean from 06:00 through 22:00,

$$\bar{p}_{id} = \frac{1}{16} \int_{06:00}^{22:00} p_i(t)\, dt.$$

The primary sample requires all 16 hours to be covered. It excludes 201 incomplete station–fuel–days across the underlying panels, including 60 German diesel days. Missing same-day quotes are carried forward from the last valid quote, without an age cap in the primary specification; 3-, 7-, and 14-day caps are reported as sensitivity checks. Quotes outside €0.50–€3.00 per liter are removed before constructing the path.

Equal timestamps with different prices are treated as ambiguous rather than resolved by file order. Germany has no such conflicts; France has three ambiguous timestamps. The affected intervals are excluded from the primary measure, and first- and last-record rules serve as sensitivities. The same rule applies to diesel, E5, and E10.

4.3 Geography and other covariates

Distance bands are defined by driving distance to the nearest station across the border: 0–10, 10–20, 20–30, 30–40, 40–50, and more than 50 km. The final category is the inland benchmark. Public- and school-holiday indicators capture predictable demand shifts. German stations are assigned to 2022 municipalities by coordinates and then, through the municipal reference table of the Federal Institute for Research on Building, Urban Affairs and Spatial Development (Bundesinstitut für Bau-, Stadt- und Raumforschung, 2022), to Eurostat's NUTS-3 urban–rural typology (Eurostat, 2026). All 2,880 German stations are classified; one station 96.7 meters outside a polygon boundary is assigned to the nearest polygon edge. Table C.3 reports the state and urban–rural composition.

Table 1. Diesel prices before and during the German tax cut

Country	Period	Mean price	Median	SD	Holiday share	N
France	Pre	1.897	1.894	0.075	0.334	36,830
France	Post	1.990	1.985	0.110	0.609	70,711
Germany	Pre	2.022	2.014	0.073	0.201	136,990
Germany	Post	1.985	1.972	0.079	0.497	264,338

Notes: Prices are euros per liter. "Pre" is 14 April–31 May 2022; "Post" is 1 June–31 August 2022. N is station-days. The German counts include 1,078 station-days without a valid routing distance; regressions exclude them. The holiday share is the mean of an indicator for a public or school holiday.

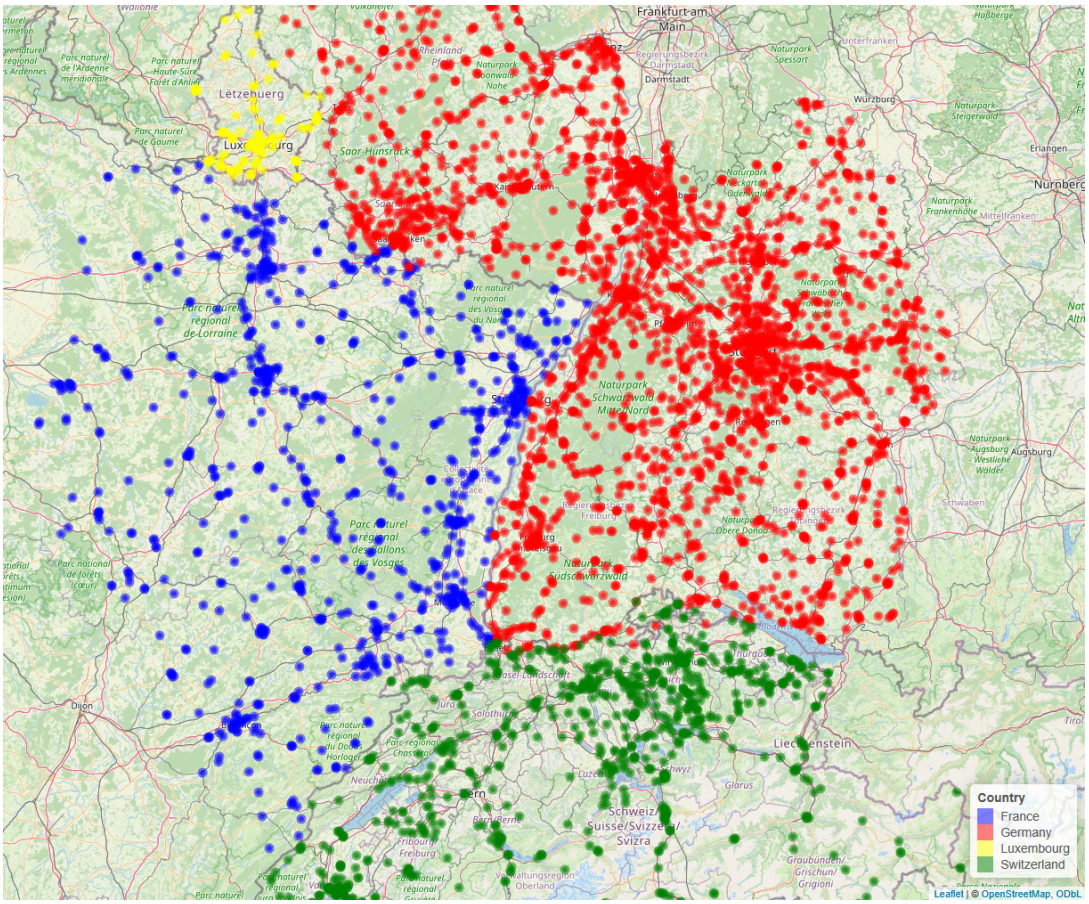


Figure 1. Fuel stations in the four-country border region

Notes: German stations are red, French stations blue, Luxembourg stations yellow, and Swiss stations green. Estimation uses German and French prices. Map lines delineate study areas and do not necessarily depict accepted national boundaries. Basemap © OpenStreetMap contributors, ODbL.

5 Empirical design

5.1 Ring specification

Following the grouping logic of spatial difference-in-differences designs (Butts, 2023), the primary regression is

$$p_{id} = \alpha_i + \mu_d + \sum_{g \neq (FR,>50)} \beta_g(\mathbf{1}\{i \in g\} \times Post_d) + H'_{id}\chi + \varepsilon_{id}. \tag{6}$$

Here p_{id} is the duration-weighted price at station i on date d, α_i and μ_d are station and date fixed effects, and H_{id} contains public- and school-holiday indicators. The eleven non-omitted groups combine country with driving-distance rings; French stations more than 50 km from Germany are the reference group. $Post_d$ equals one from 1 June through 31 August 2022. Standard errors are clustered by station in the primary specification.

For a German group, β_g is its post-policy price change relative to inland France, conditional on the fixed effects and holiday controls. I report diesel pass-through as $-\beta_g/0.167076$ and gasoline pass-through as $-\beta_g/0.351645$. For French rings, no German tax was remitted; the same normalization is only a tax-equivalent scale for comparing magnitudes and is not literal French tax pass-

through. Because German stations and inland French stations do not follow parallel pre-reform trends (Section 7), these rates describe the German response relative to inland France; they are not identified pass-through levels. The article's identified object is the within-Germany contrast that follows.

The coefficient difference $\beta_{DE,0\text{–}10} - \beta_{DE,>50}$ isolates the additional near-border response relative to inland German stations. It removes any France-wide counterfactual component common to both German rings. Under parallel untreated trends across German distance groups, it is the article's principal spatial contrast. This within-Germany comparison is also insulated from France-wide movements that could otherwise contaminate cross-country estimates, an important concern in contemporaneous evidence on the reform (Montag et al., 2023).

5.2 Continuous and dynamic specifications

Restricted cubic splines with knots at 10, 20, 30, 40, and 50 km provide a continuous summary of the German distance gradient. Weekly interactions trace adjustment during the policy. Event-study interactions compare the 0–10 km German ring with German stations more than 50 km from France. Event time is measured in seven-day intervals: 1–7 June is week 0, and 25–31 May (week -1) is omitted. Week 11 pools 17–31 August. An anticipation sensitivity instead omits week -2 and estimates the final pre-policy week; cutoff sensitivities replace 10 km with 8 or 12 km. These specifications test whether the ring result reflects a sharp bin choice, whether spatial differences predate the reform, and whether demand postponement immediately before 1 June affects the comparison.

Identification requires that, absent the German cut, the included groups would have followed the omitted inland-French trend after conditioning on fixed effects and holidays. The German near-versus-inland contrast requires the weaker assumption of parallel trends across German distance groups. Cross-border commuting, tourism, local demand, procurement, or other shocks that change differentially at the border can violate these assumptions. Section 7 evaluates these possibilities with daily and weekly event studies, bounds on pre-trend violations, a reversal test at the expiry of the cut, controls for station composition and county-specific shocks, and inference that allows for correlation within counties and across space.

Figure 2 shows the aggregate price paths that motivate the design. German prices fall sharply when the tax cut begins, and French prices also move over the policy period. The final weeks precede the expiration of the German cut and the increase in the French rebate on 1 September. These national series establish timing and scale, not the spatial counterfactual: identification comes from differences across the pre-specified distance groups.

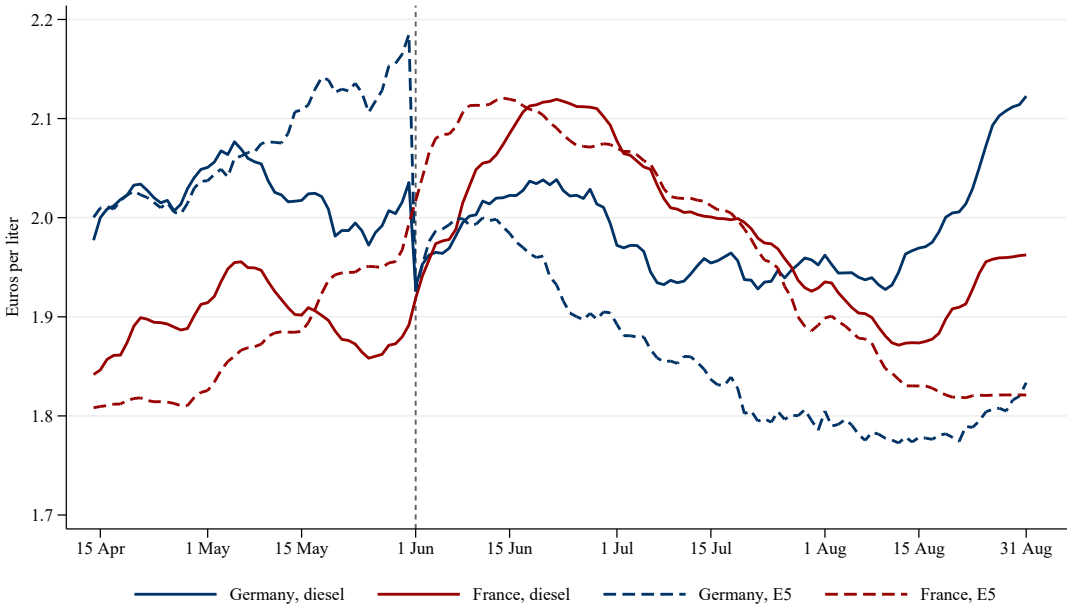


Figure 2. French and German country-average diesel and E5 prices

Notes: Daily averages across stations in each country, in euros per liter. The vertical line marks 1 June 2022, the first day of the German tax cut.

6 Results

6.1 Diesel: the border premium

Table 2 reports the primary ring estimates. German diesel prices fall by 14.18 cents per liter within 10 km of France, equivalent to 84.9% pass-through, compared with 13.22 cents, or 79.1%, more than 50 km inland. The near-border premium is 0.96 cents per liter (95% CI: 0.43 to 1.49; $p < 0.001$), or 5.8 percentage points of the tax-inclusive cut. This economically meaningful local difference is the article's central result. Responses elsewhere are also heterogeneous: the 20–30 and 30–40 km rings exceed the inland response, whereas the 10–20 and 40–50 km estimates are closer to it. Cross-border exposure is therefore important but not summarized by a monotone gradient.

French prices near the border also fall. Relative to French stations more than 50 km inland, diesel prices fall by 0.81 cents in the 0–10 km ring, 1.14 cents in the 10–20 km ring, and 0.73 cents in the 20–30 km ring; the 30–40 km estimate is -0.87 cents ($p = 0.082$), and only the 40–50 km estimate is positive. This is the sign the model predicts for an untreated rival. French border prices were already falling relative to inland France in the second half of May, however (Section 7.1), so I do not interpret the magnitudes of the French estimates; below they serve only as an upper bound on the German-caused French response.

The German premium is a net competition effect, not evidence that foreign feedback is strong. With the French 0–10 km estimate as the foreign response, the bound in Section 3 limits feedback through French prices to 0.3–0.5 of the 5.8 percentage points. The remainder must reflect how border and inland markets differ in absorbing the cut; Section 6.2 shows that the premium arises where the cut closed the cross-border price gap. The nonmonotone ring profile fits this interpretation, because

treated domestic competition and the mass of marginal cross-border shoppers both vary across local markets. Figure 3 displays the diesel estimates on both sides of the border.

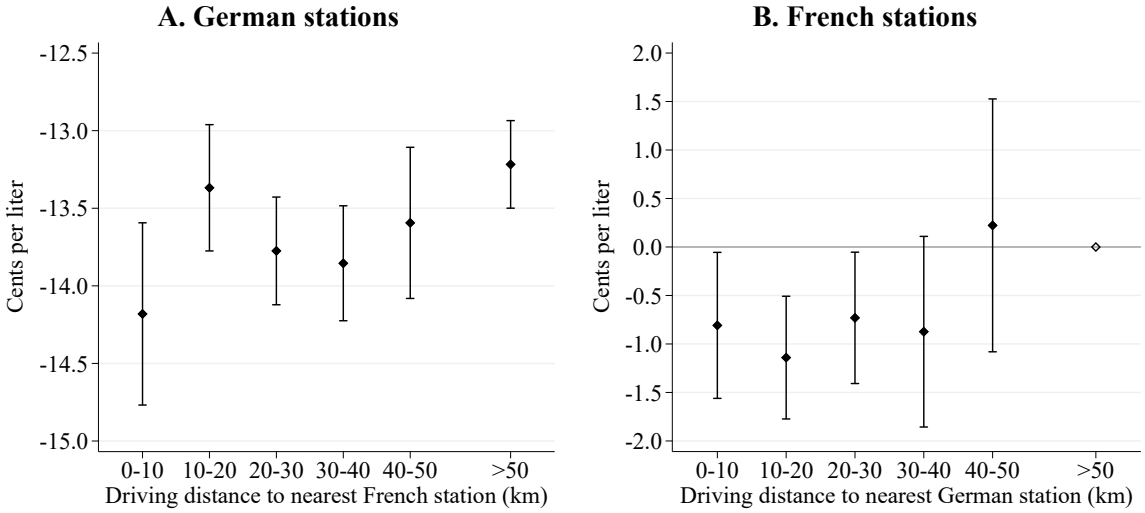


Figure 3. Diesel price responses by distance on both sides of the border

Notes: Points are the Table 2 coefficients in cents per liter; bars are 95% confidence intervals based on station-clustered standard errors. Both panels are relative to French stations more than 50 km from Germany, the omitted group shown as the hollow marker in Panel B. Vertical scales differ across panels.

Table 2. Spatial price responses to the German fuel-tax cut: diesel

Country and distance	Price change (cents/liter)	SE	*p*	Tax-equivalent response (%)
Germany, >50 km	-13.22	0.14	<0.001	79
Germany, 0–10 km	-14.18	0.30	<0.001	85
Germany, 10–20 km	-13.37	0.21	<0.001	80
Germany, 20–30 km	-13.77	0.18	<0.001	82
Germany, 30–40 km	-13.85	0.19	<0.001	83
Germany, 40–50 km	-13.59	0.25	<0.001	81
France, 0–10 km	-0.81	0.38	0.035	5
France, 10–20 km	-1.14	0.32	<0.001	7
France, 20–30 km	-0.73	0.35	0.035	4
France, 30–40 km	-0.87	0.50	0.082	5
France, 40–50 km	0.22	0.67	0.738	-1

Notes: The dependent variable is reported in cents per liter. The omitted group is French stations more than 50 km from Germany. The model includes station and date fixed effects plus public- and school-holiday controls. Standard errors are clustered by station. The final column divides the negative coefficient by the 16.7076-cent tax-inclusive diesel cut; for French rings it is a tax-equivalent response, not French tax pass-through. N = 507,791.

6.2 Gasoline and continuous distance

German gasoline pass-through is high across the distance distribution. For E5, it is 89.7% in the nearest ring and 91.5% inland; for E10, 88.9% and 90.8%. Thus the nearest-ring responses are 1.8 and 1.9 percentage points lower, respectively, and neither difference is statistically significant ($p = 0.197$ for E5; $p = 0.176$ for E10). The diesel premium therefore does not arise mechanically from proximity. The fuels differ in how the cut moved the cross-border price gap. Germany was

initially more expensive than France by about 12.5 cents for diesel, 19.8 cents for E5, and 19.0 cents for E10. During the reform, the diesel gap nearly closed, whereas the gasoline gaps reversed to roughly -11 to -13 cents (Figures B.1 and B.2). In the terms of Section 2.2, the cut appears to have made cross-border competition active for diesel, because German and French diesel became close substitutes for motorists near the border. For gasoline the gap reversed but stayed wide: Germany went from much dearer to much cheaper than France, so French stations did not become a close alternative for German motorists, and competition across the border stayed inactive. Within the policy period, week-to-week fluctuations in the diesel gap do not predict the size of the premium, so the evidence for this account comes from the contrast across fuels and from the end of the cut (Section 7.2), not from weekly variation.

Several French gasoline coefficients are negative, but their spatial ordering is uneven. For E5, the 10–20 and 30–40 km coefficients are significant at 5%, while the 0–10 km estimate narrowly misses that threshold. For E10, the 0–10, 10–20, and 20–30 km coefficients are significant. As for diesel, I do not interpret their magnitudes (Tables C.4 and C.5).

Restricted cubic splines summarize rather than resolve this heterogeneity. Diesel has a distinct near-border peak; the gasoline curves are nonmonotone and generally rise away from the border. Because spline coefficients are basis-dependent, the estimated curves—not individual basis terms—are the economically interpretable objects. Figure B.3 places the three curves on one page.

6.3 Evolution over the policy period

Pass-through varies substantially over time. Weekly diesel estimates are high during the early and middle policy weeks, and the distance rings comove closely (Figure 4). Most temporal variation is therefore national, with a smaller spatial differential superimposed on the common adjustment path. The final weeks precede the expiry of the cut; Section 7.2 examines the expiry itself. Related evidence likewise documents pronounced time variation in average pass-through (Freitas and Syga, 2022; Dovern et al., 2023; Frondel et al., 2026).

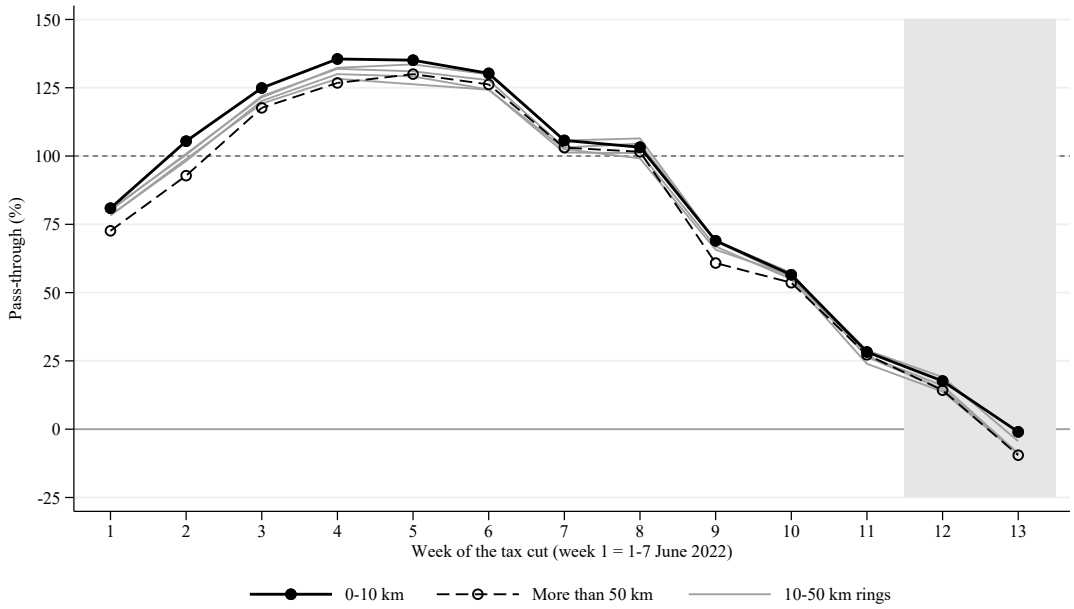


Figure 4. Weekly diesel pass-through by German distance ring

Notes: Each series is estimated separately for every policy week from the pre-policy period and that week. It is the price change of one German distance ring relative to all French stations, with station and date fixed effects and holiday controls, divided by the 16.71-cent tax-inclusive diesel cut. Week 13 runs from 24 to 31 August. The shaded weeks (17–31 August) precede the expiry of the German cut and the increase in the French rebate on 1 September and are not interpreted.

7 Identification and robustness

7.1 Pre-trends and timing

Figure 5 compares the 0–10 km German diesel ring with German stations more than 50 km inland, using the final pre-policy week as the omitted period. The differential turns negative exactly when the tax cut begins, equals -1.47 cents in week 0 and -1.98 cents in week 1, and remains negative throughout the policy period. Ten of twelve post-reform coefficients are individually significant at 5%, and they are jointly significant ($p < 0.001$). The same profile appears with 8 km and 12 km cutoffs (Figure B.4).

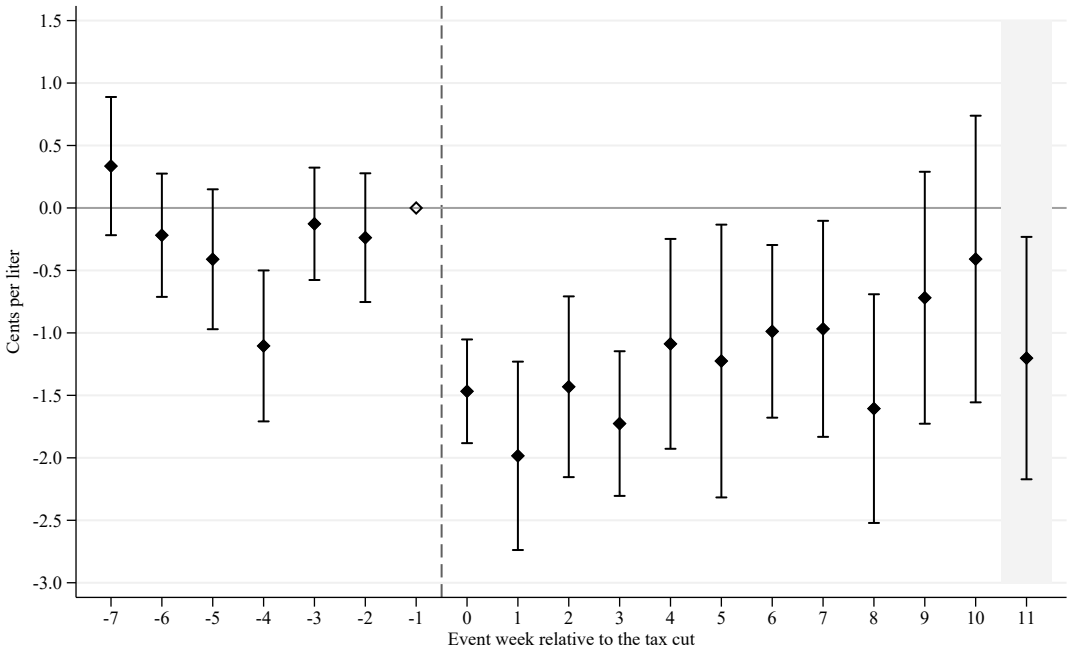


Figure 5. Event study: German 0–10 km diesel ring relative to inland Germany

Notes: Coefficients in cents per liter compare German stations within 10 km of a French station with German stations more than 50 km away. Event week 0 is 1–7 June 2022; the shaded week 11 pools 17–31 August. The omitted week −1 (25–31 May) is shown as a hollow marker. Bars are 95% confidence intervals based on station-clustered standard errors. Five of six leads are individually insignificant; the six leads are jointly significant ($F = 11.05$, $p < 0.001$).

The pre-period evidence calls for precision. Five of six leads are individually insignificant; the exception is week -4, at -1.10 cents. The six leads are jointly significant ($F = 11.05$, $p < 0.001$), and the other five still reject jointly when week -4 is left out ($p < 0.001$), so strict parallel pre-trends are rejected. The leads do not drift toward the post-reform level, however: a linear pre-trend is -0.01 cents per week (SE 0.06). Bounds that allow post-reform violations of parallel trends (Rambachan and Roth, 2023; Roth, 2022) make the distinction precise. If post-reform violations may be as large as the largest pre-reform violation (relative magnitude $\bar{M} = 1$), the 95% interval for the first policy week is -2.89 to -0.02 cents. For the average over the policy period, the robust interval includes zero as soon as any violation is allowed. The break at the start of the cut is therefore robust, whereas the full-period average of 0.96 cents is not an assumption-free parameter.

Daily data locate the break. With 31 May omitted, the near-border coefficient on 1 June is -1.35 cents (SE 0.47); relative to the mean of the 20 preceding days it is -1.95 cents (SE 0.37) (Figure B.6). The daily leads also reject jointly: most pre-reform days lie 0.5–1.2 cents above 31 May, so the last day before the cut was itself low, which is why the comparison with the 20-day mean is the more informative one. Re-normalizing the weekly event study to week -2 gives a final-week coefficient of 0.24 cents ($p = 0.366$), so the break is not an artefact of the omitted week or of last-week demand postponement (Figure B.5). Assigning a false treatment date of 18 May yields +0.61 cents for the nearest German ring ($p = 0.050$), opposite in sign to the reform-period premium. The same placebo gives negative coefficients of 1.0 to 1.9 cents for four French rings (Table C.7): French border prices were already falling relative to inland France before the reform, which is why the French magnitudes are not interpreted.

7.2 Reversal when the tax returns

If the premium reflects how border stations pass through German tax changes, it should reverse when the tax returns. On 1 September 2022 the German tax rose back by 16.71 cents per liter for diesel and 35.16 cents for gasoline. Two French changes on the same day work against a reversal: the French rebate rose from 18 to 30 cents, and TotalEnergies added its own rebate of 20 cents. Lower French prices give German border stations a reason to raise prices by less than inland stations, so the French changes are unlikely to produce a reversal of the premium. The sample ends on 26 September, before the French refinery strikes.

Figure 6 extends the event study to 26 September. The diesel premium persists through the first week of September (-1.05 cents) and then fades: it is -0.51 cents in the second week and +0.29 and +0.41 cents in the third and fourth. Over 1–26 September it averages 0.01 cents, against -0.98 cents during the cut; the change of +0.99 cents has a station-clustered p-value of 0.053. In symmetric 26-day windows, the near-border diesel price falls by 1.29 cents relative to inland when the cut begins and rises by 0.75 cents when it ends; the sum of the two, -0.55 cents, is not distinguishable from zero ($p = 0.21$). The premium is therefore tied to the policy period. It disappears once German diesel again became much dearer than French diesel, in line with the active-competition account.

Gasoline shows the mirror image. Near-border stations cut E5 prices by 0.57 cents less than inland stations in June and raised them by 1.07 cents less in September, so they passed through slightly less of both the cut and its reversal; dropping motorway stations or stations near Luxembourg, Switzerland, and Belgium leaves both patterns intact.

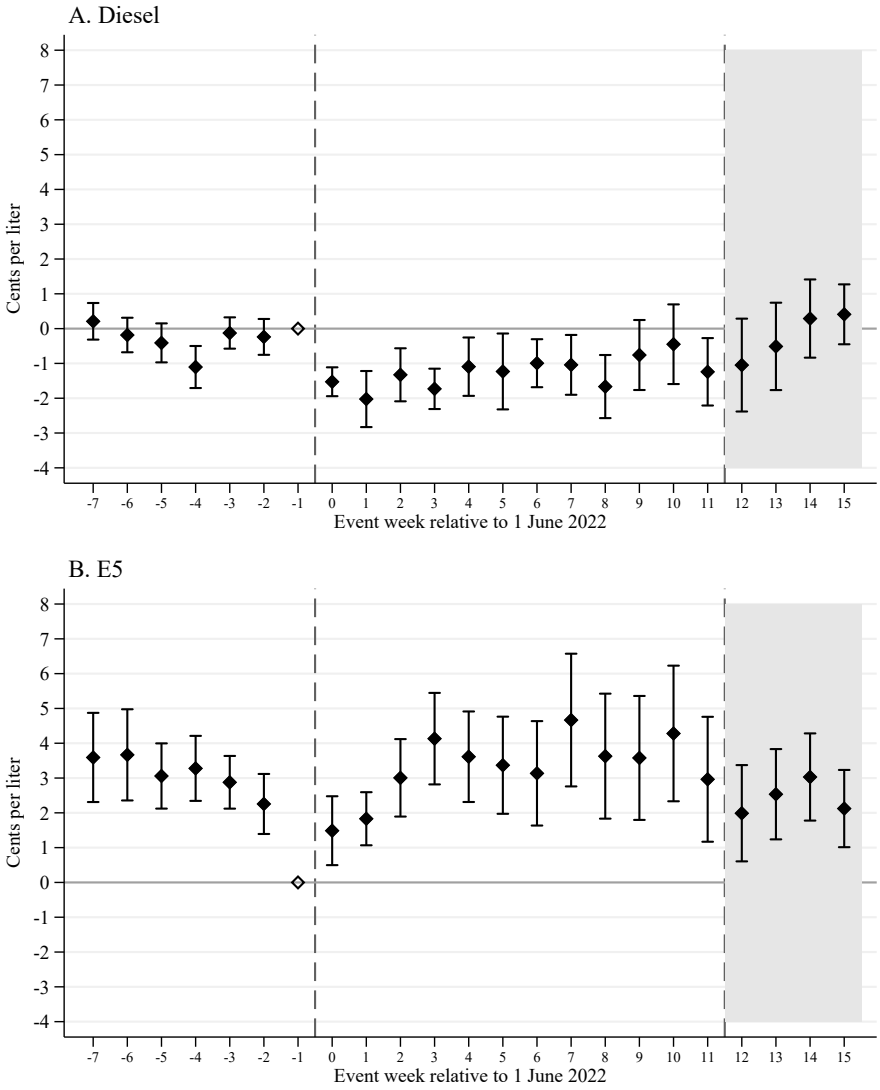


Figure 6. Event study through the expiry of the cut: German stations within 10 km of France relative to inland Germany

Notes: Coefficients in cents per liter for diesel (Panel A) and E5 (Panel B). Event week 0 is 1–7 June 2022; week 11 pools 17–31 August; weeks 12–15 (shaded) cover 1–26 September, after the German tax returned. The omitted week −1 (25–31 May) is shown as a hollow marker; dashed lines mark 1 June and 1 September. Station and date fixed effects and holiday controls; bars are 95% confidence intervals based on station-clustered standard errors. N = 380,665 station-days and 2,309 stations (diesel); N = 377,013 station-days and 2,287 stations (E5).

7.3 Composition, local shocks, and distance

Table C.6 reports the premium under alternative specifications in the German-only sample, where the baseline is -0.97 cents. Allowing the post-reform change to vary with the pre-reform price, the five largest brands, motorway location, urban–rural type, and the number of German rivals within 5 km reduces it to -0.76 cents. County-by-date fixed effects, which absorb every shock common to a county, give -0.79 cents; the coarser fixed effects for government districts (*Regierungsbezirke*) by date give -1.31 cents. Dropping motorway stations gives -0.95 cents, dropping stations within 20 km of Luxembourg, Switzerland, or Belgium gives -0.83 cents, and ending the policy period on

16 August gives -0.97 cents. Against stations 10–50 km from France instead of inland stations, the premium is -0.54 cents ($p = 0.050$).

The premium is concentrated close to the border. With near-border cutoffs of 4, 6, 8, 10, 12, 15, and 20 km, it equals -1.70, -0.98, -0.93, -0.97, -1.07, -0.67, and -0.37 cents; driving-time cutoffs give the same profile. A triple difference that compares diesel with E5 at the same stations, and so absorbs every station-by-date shock common to both fuels, gives -1.58 cents (station-clustered SE 0.51), or -7.5 percentage points of pass-through.

7.4 Inference

The near-border premium is estimated with station-clustered standard errors throughout, matching the station-level assignment of distance to France. Fuel prices can also move together within counties, so Table C.8 reports alternatives for the German-only premium of -0.97 cents. Clustering by the 86 counties roughly doubles the standard error, to 0.54 cents ($p = 0.078$), and a county wild cluster bootstrap gives $p = 0.088$; spatial-HAC standard errors (Conley, 1999) with cutoffs between 10 and 100 km give p-values between 0.05 and 0.18. The premium is therefore significant at conventional levels with station clustering and at the 10% level with county clustering. The September change in Section 7.2 is less precisely estimated with county clustering ($p = 0.38$).

7.5 Price construction

Results survive reasonable price-construction variants. Using the full day yields 83.5% pass-through in the nearest German ring and 78.0% inland. Limiting carried quotes to 3, 7, or 14 days raises levels somewhat but retains the nearest-ring differential. First-record and last-record rules at ambiguous French timestamps leave diesel and E5 unchanged and affect only three E10 station-days. Tables C.9 and C.10 report the sensitivities and panel construction.

7.6 Scope

The data measure posted prices, not quantities, margins, station-level wholesale costs, or consumer trips. The design therefore cannot separate cross-border shopping from correlated local demand, procurement, or market-structure changes. Driving distance is measured precisely, but the nearest foreign station need not be every motorist's relevant alternative. The active-competition account rests on the contrast across fuels and on the end of the cut; quantity or travel data would be needed to measure cross-border switching directly. The article identifies spatial price responses and assesses whether they comport with competition; it does not estimate welfare or a structural demand system.

8 Conclusion

National average incidence conceals a local response to Germany's 2022 fuel-tax cut. German diesel stations within 10 km of a French station cut prices by 0.96 cents per liter more than stations more

than 50 km inland, 5.8 percentage points of the tax-inclusive cut. The differential appears on the first day of the cut, is largest within a few kilometers of the border, survives controls for station composition and county-specific shocks, and disappears after the tax returns. Gasoline, for which the cut made Germany much cheaper than France, shows no such premium. The jointly rejected pre-trend test and the wider confidence intervals under county clustering counsel against treating 0.96 cents as a precise structural parameter, but the timing, the distance gradient, and the reversal tie the differential to the cut.

A model with a treated border station, a treated domestic rival, and an untreated foreign rival shows that, at a given intensity of cross-border substitution, feedback through foreign prices affects German pass-through only at second order and, given the size of the French response, explains at most a small part of the premium. The pattern across fuels and over time is consistent with a change in that intensity. Where the cut brought German and French prices close together, as for diesel, competition across the border became active and border stations passed through more of the cut. Where the gap stayed wide, as for gasoline, or where the end of the cut reopened it, as for diesel in September, competition was inactive and the premium was absent.

For tax policy, a national fuel-tax change reaches consumers unevenly within the country, and the variation depends on the fuel and on how the change moves the cross-border price gap. Evaluations that average over all stations miss it. Because the same tax change can switch cross-border competition on for one product and leave it off for another, the incidence of a unilateral tax change at a border depends on the neighbor's prices as well as on the domestic market.

A Model derivations

Dividing the firms' first-order conditions by b yields equation (2). Strict own-price concavity follows immediately because

$$\frac{\partial^2 \pi_j}{\partial p_j^2} = -2b < 0 \qquad (j \in \{G, D, F\}).$$

The coefficient matrix of the price system is

$$A = \begin{pmatrix} 2 & -\delta_B & -\rho \\ -\delta_B & 2 & 0 \\ -\rho & 0 & 2 \end{pmatrix}, \qquad \det(A) = 2(4 - \delta_B^2 - \rho^2).$$

Thus a unique interior equilibrium exists under the maintained stability condition

$$\Delta_B \equiv 4 - \delta_B^2 - \rho^2 > 0, \tag{A.1}$$

together with positive equilibrium quantities. A sufficient primitive restriction is $\delta_B, \rho \in [0, 1)$, which makes the best-response map a contraction. Integrable demand with strictly concave quasi-linear utility requires the stronger condition $\delta_B^2 + \rho^2 < 1$, under which the matrix of demand slopes in equation (1) is negative definite; this condition implies $\Delta_B > 0$. Nonsingularity alone requires $\Delta_B \neq 0$; the stronger condition (A.1) fixes the economically relevant signs of the comparative statics. Symmetry of the off-diagonal *G–F* terms follows from the integrable local demand system in equation (1). In the two-firm special case, restoring the unnormalized cross-price slope $\gamma = b\rho$ gives the first-order-condition matrix

$$\begin{pmatrix} 2b & -\gamma \\ -\gamma & 2b \end{pmatrix},$$

whose determinant is $4b^2 - \gamma^2$. The expression is therefore the determinant condition for a unique local price equilibrium, not an additional behavioral assumption.

To derive pass-through, let $g = dp_G/d\tau$, $d = dp_D/d\tau$, and $f = dp_F/d\tau$, where $d\tau$ raises both German marginal costs by one unit. Differentiating the first-order conditions gives

$$\begin{aligned} 2g - \delta_B d - \rho f &= 1, \\ 2d - \delta_B g &= 1, \\ 2f - \rho g &= 0. \end{aligned} \tag{A.2}$$

The second and third equations imply $d = (1 + \delta_B g)/2$ and $f = \rho g/2$. Substitution into the first gives

$$g = \frac{2 + \delta_B}{4 - \delta_B^2 - \rho^2}, \qquad f = \frac{\rho}{2} g,$$

which proves equation (3). For an inland German station and its treated rival, the derivative system is $2i - \delta_I j = 1$ and $2j - \delta_I i = 1$. Symmetry gives $i = j = 1/(2 - \delta_I)$, proving (4).

Finally, $\Pi_B > \Pi_I$ is equivalent, under positive denominators, to

$$\begin{aligned}(2 + \delta_B)(2 - \delta_I) &> 4 - \delta_B^2 - \rho^2,\\ \rho^2 &> (2 + \delta_B)(\delta_I - \delta_B),\end{aligned}$$

which is condition (5). The foreign response $f = (\rho/2)g$ is linear in ρ around $\rho = 0$, whereas

$$\frac{\partial \Pi_B}{\partial \rho} = \frac{2\rho(2 + \delta_B)}{(4 - \delta_B^2 - \rho^2)^2}$$

equals zero at $\rho = 0$. Hence foreign feedback affects German pass-through only at second order near zero. If $\delta_B = \delta_I$, condition (5) reduces to $\rho^2 > 0$; if $\rho = 0$, $\Pi_B = 1/(2 - \delta_B)$, so the premium has the sign of $\delta_B - \delta_I$.

Finally, equation (3) differentiates an equilibrium at fixed slopes. If ρ varies with the price gap z, the total derivative is $dp_G/d\tau = \Pi_B + (\partial p_G/\partial \rho)\, \rho'(z)\, dz/d\tau$. In a smooth heterogeneous-choice model the cross-price slope is governed by the density of motorists at the switching threshold, so $\rho'(z)$ can take either sign and the additional term depends on markups and demand curvature; the linear model does not sign it.

B Additional figures

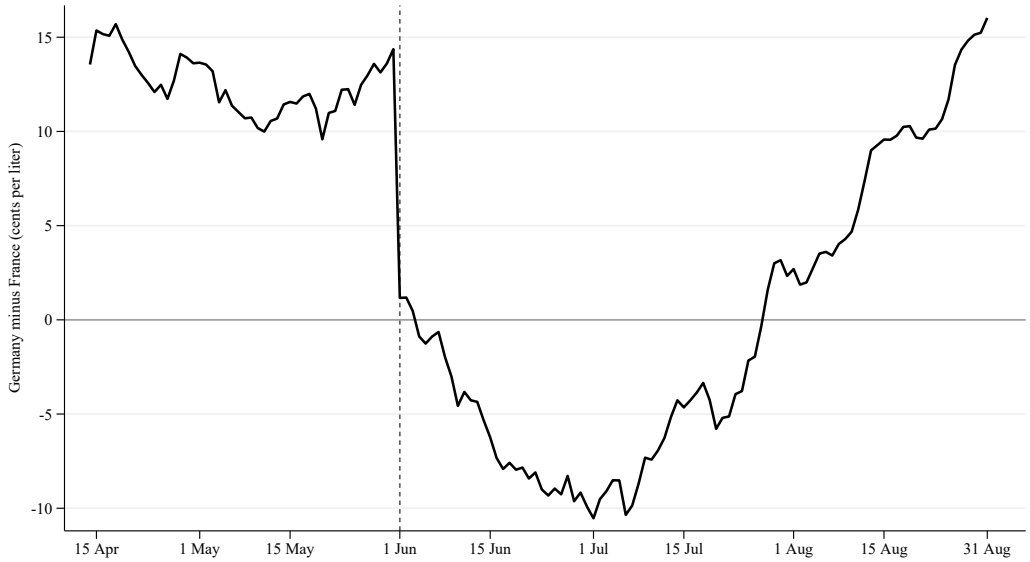


Figure B.1. Diesel-price difference between Germany and France

Notes: Daily German country-average price minus French country-average price, in cents per liter. The vertical line marks 1 June 2022.

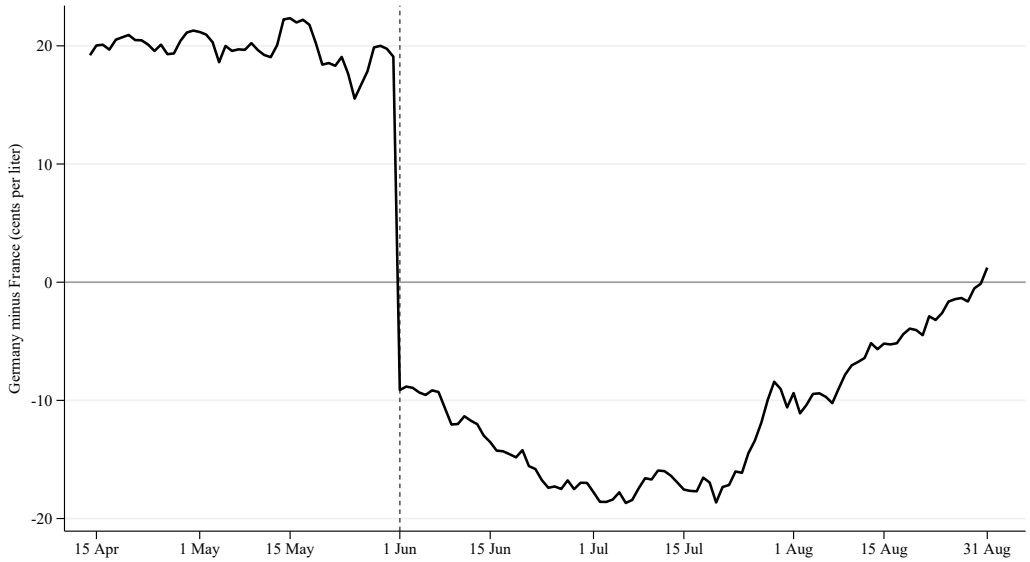


Figure B.2. E5-price difference between Germany and France

Notes: Daily German country-average price minus French country-average price, in cents per liter. The vertical line marks 1 June 2022.

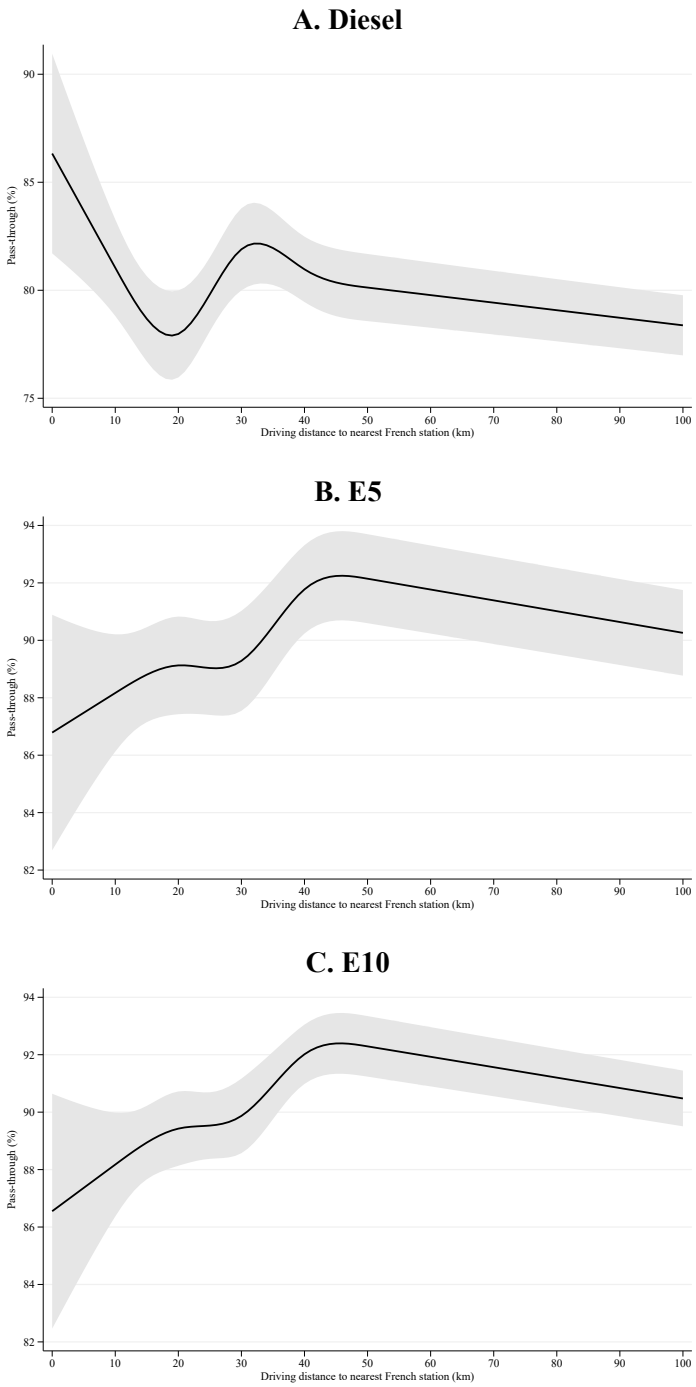


Figure B.3. Predicted German pass-through versus distance by fuel

Notes: Restricted cubic splines use knots at 10, 20, 30, 40, and 50 km. Diesel exhibits a distinct near-border peak; the gasoline curves are nonmonotone and generally rise away from the border. The plotted curves, rather than individual basis coefficients, are the economically interpretable objects.

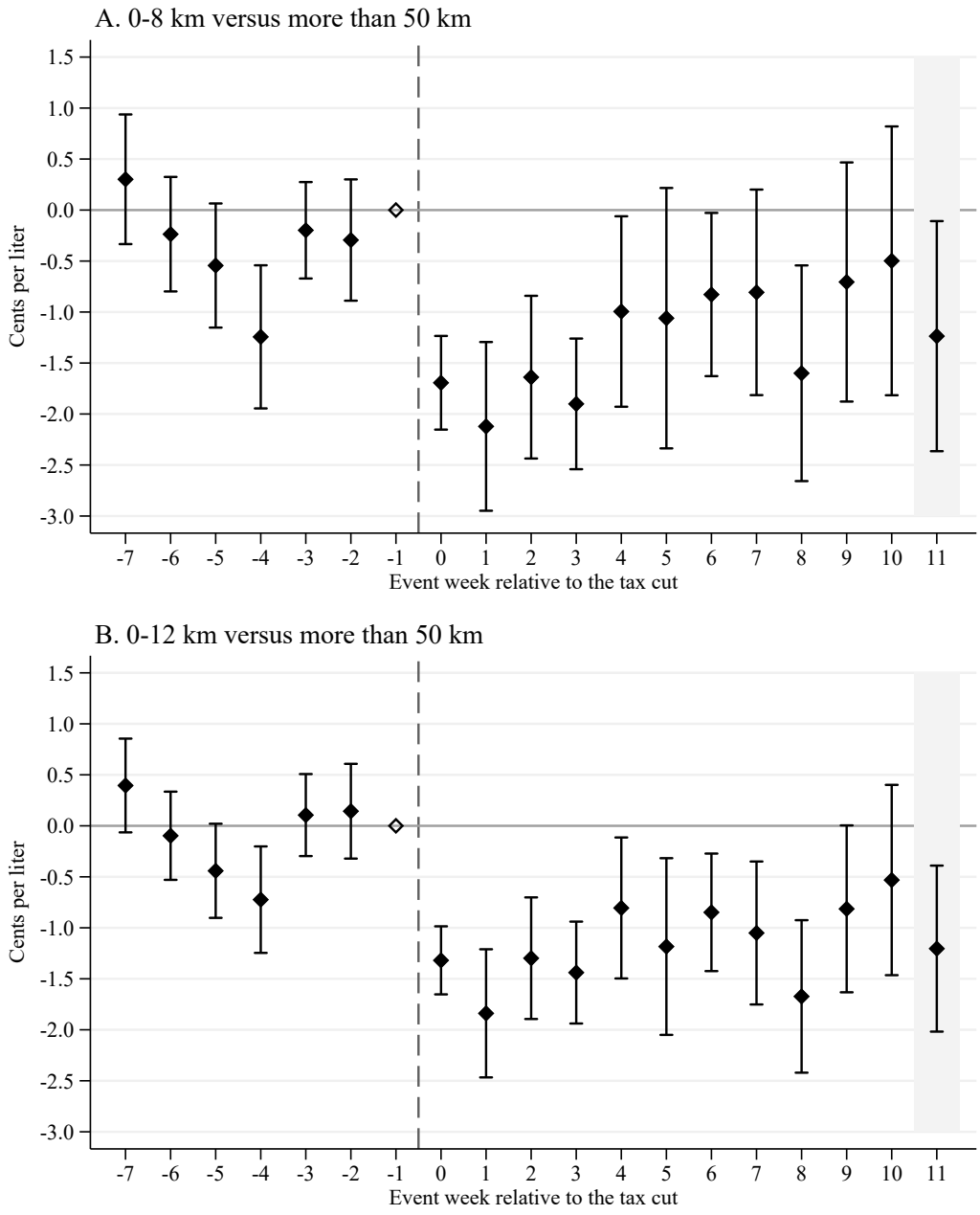


Figure B.4. Event-study sensitivity to the near-border cutoff: German diesel

Notes: Coefficients in cents per liter. Panels replace the primary 0–10 km group with 0–8 km and 0–12 km groups; the comparison group remains German stations more than 50 km from France. Event week −1 (hollow marker) is omitted; the shaded week 11 pools 17–31 August. The near groups contain 53, 67, and 90 stations at the 8, 10, and 12 km cutoffs, respectively. All three specifications reject the joint null for the six pre-policy coefficients and for the post-policy coefficients.

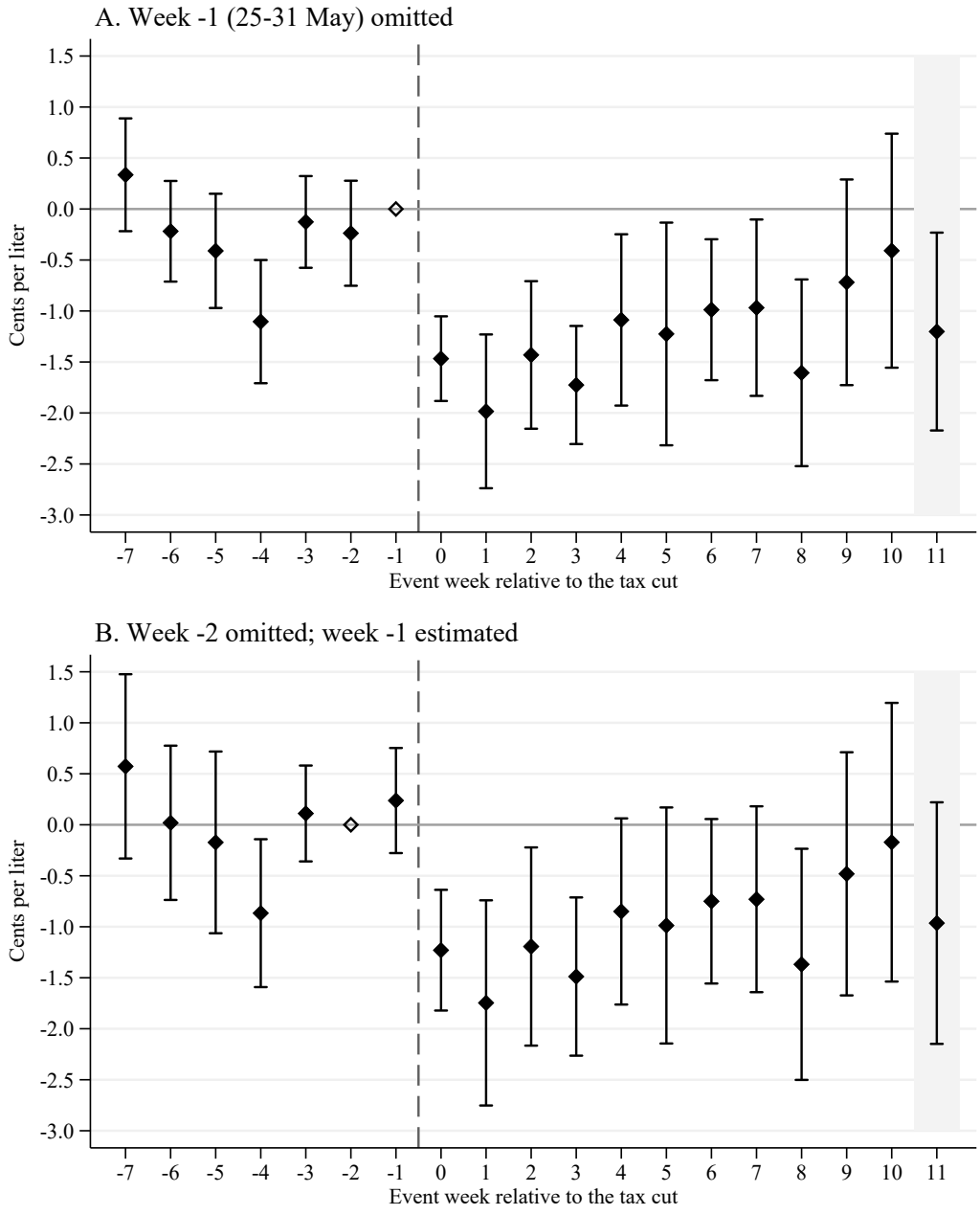


Figure B.5. Event-study normalization and anticipation sensitivity: German diesel

Notes: Coefficients in cents per liter. Panel A omits event week −1 (25–31 May), as in Figure 5. Panel B omits week −2 and estimates the final pre-policy week explicitly; its week −1 coefficient is 0.24 cents (SE 0.26; $p = 0.366$), providing no evidence of a differential last-week anticipation effect. Hollow markers show the omitted week. Re-normalization changes the reference point, not fitted values.

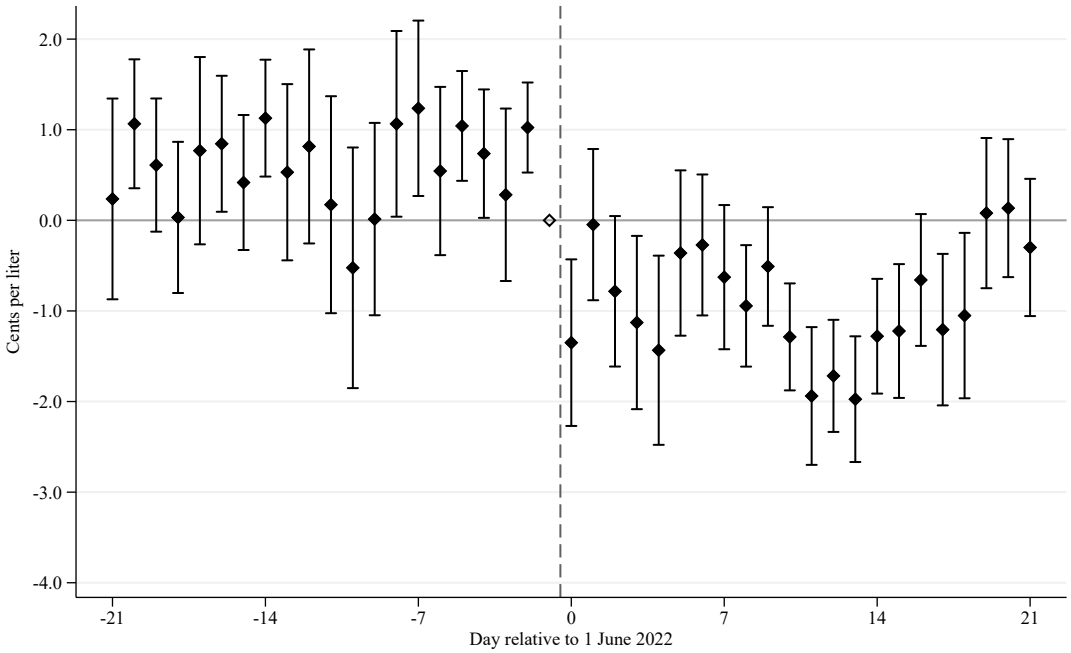


Figure B.6. Daily event study: German 0–10 km diesel ring relative to inland Germany

Notes: Coefficients in cents per liter for days -21 to +21 around 1 June 2022 (day 0); 31 May (day -1, hollow marker) is omitted. Station and date fixed effects and holiday controls; bars are 95% confidence intervals based on station-clustered standard errors. N = 98,218 station-days, 2,285 stations. Relative to the mean of days -21 to -2, the day-0 coefficient is -1.95 cents (SE 0.37).

C Additional tables

Table C.1. E5 descriptive statistics

Country	Period	Mean price	Median	SD	Holiday share	N
France	Pre	1.874	1.869	0.083	0.340	13,757
France	Post	1.973	1.988	0.146	0.613	26,816
Germany	Pre	2.073	2.062	0.082	0.201	135,455
Germany	Post	1.859	1.844	0.106	0.497	261,296

Notes: Prices are euros per liter; N is station-days. Periods as in Table 1.

Table C.2. E10 descriptive statistics

Country	Period	Mean price	Median	SD	Holiday share	N
France	Pre	1.827	1.809	0.093	0.332	28,103
France	Post	1.929	1.937	0.136	0.610	54,399
Germany	Pre	2.017	2.006	0.087	0.200	131,402
Germany	Post	1.803	1.787	0.112	0.497	253,548

Notes: Prices are euros per liter; N is station-days. Periods as in Table 1.

Table C.3. German stations by state and Eurostat urban–rural typology

Panel A. Federal states	
Baden-Württemberg	1,940
Rheinland-Pfalz	773
Saarland	167
Panel B. Eurostat NUTS-3 urban–rural typology	
Predominantly urban	1,022
Intermediate	1,263
Predominantly rural	595
Total	2,880

Notes: Stations are assigned to 2022 municipalities with the VG5000 geometry and then, through the BBSR municipal reference table, to Eurostat's NUTS-3 typology. All 2,880 stations are classified. One station 96.7 meters outside a polygon boundary is assigned to the nearest polygon edge; no station has multiple interior matches.

Table C.4. Spatial price responses: E5

Country and distance	Price change (cents/liter)	SE	p	Tax-equivalent response (%)
Germany, >50 km	-32.18	0.29	<0.001	92
Germany, 0–10 km	-31.55	0.56	<0.001	90
Germany, 10–20 km	-31.95	0.32	<0.001	91
Germany, 20–30 km	-32.23	0.33	<0.001	92
Germany, 30–40 km	-32.43	0.35	<0.001	92
Germany, 40–50 km	-31.89	0.37	<0.001	91
France, 0–10 km	-2.95	1.52	0.052	8
France, 10–20 km	-2.28	0.73	0.002	6
France, 20–30 km	-1.91	1.12	0.088	5
France, 30–40 km	-2.76	1.14	0.015	8
France, 40–50 km	-1.87	1.71	0.274	5

Notes: Specification as in Table 2. The final column divides the negative coefficient by the 35.16-cent tax-inclusive gasoline cut; for French rings it is a tax-equivalent scale, not French tax pass-through. N = 437,324. Equality of Germany 0–10 km and Germany more than 50 km: $p = 0.197$.

Table C.5. Spatial price responses: E10

Country and distance	Price change (cents/liter)	SE	p	Tax-equivalent response (%)
Germany, >50 km	-31.94	0.22	<0.001	91
Germany, 0–10 km	-31.26	0.54	<0.001	89
Germany, 10–20 km	-31.63	0.29	<0.001	90
Germany, 20–30 km	-32.08	0.26	<0.001	91
Germany, 30–40 km	-32.23	0.29	<0.001	92
Germany, 40–50 km	-31.69	0.32	<0.001	90
France, 0–10 km	-1.16	0.56	0.039	3
France, 10–20 km	-2.19	0.53	<0.001	6
France, 20–30 km	-1.00	0.40	0.013	3
France, 30–40 km	-0.82	0.67	0.226	2
France, 40–50 km	0.51	1.00	0.611	-1

Notes: Specification as in Table 2. The final column divides the negative coefficient by the 35.16-cent tax-inclusive gasoline cut. N = 467,452. Equality of Germany 0–10 km and Germany more than 50 km: $p = 0.176$.

Table C.6. Robustness of the German diesel premium

Specification	Cents	Station SE	County SE	Near stations
Panel A. Specification				
Baseline (German stations only)	-0.97	0.27	0.54	67
Moderators × policy period	-0.76	0.24	0.60	65
County-by-date fixed effects	-0.79	0.38	0.50	67
Government-district-by-date fixed effects	-1.31	0.31	0.60	67
Without motorway stations	-0.95	0.28	0.58	65
Without stations near other borders	-0.83	0.28	0.56	60
Policy period ending 16 August	-0.97	0.27	0.51	67
Comparison group 10–50 km	-0.54	0.28	0.52	67
Panel B. Near-border cutoff (driving distance)				
4 km	-1.70	0.51	0.69	19
6 km	-0.98	0.35	0.42	34
8 km	-0.93	0.30	0.54	53
10 km	-0.97	0.27	0.54	67
12 km	-1.07	0.24	0.52	90
15 km	-0.67	0.19	0.51	137
20 km	-0.37	0.14	0.47	238
Panel C. Triple difference, diesel minus gasoline				
Diesel minus E5	-1.58	0.51	1.34	67
Diesel minus E10	-1.65	0.52	1.38	67

Notes: Each row reports the post-policy price change of German stations within 10 km (Panel B: the stated cutoff) of a French station minus that of German stations more than 50 km away, in cents per liter. All rows include station and date fixed effects and holiday controls. The moderators row interacts the policy period with the pre-reform price, the five largest brands, a motorway indicator, the Eurostat urban–rural type, and the log number of German stations within 5 km; two near stations without a pre-reform price drop out. Government districts are *Regierungsbezirke*. The motorway row is the Table 2 contrast after dropping 37 German motorway stations. The row without other borders drops German stations within 20 km of Luxembourg or Switzerland (driving distance) or Belgium (straight-line). Panel C uses station-by-fuel, station-by-date, and fuel-by-date fixed effects. County standard errors cluster by the 86 NUTS-3 counties.

Table C.7. Placebo treatment beginning 18 May: diesel

Country and distance	Price change (cents/liter)	SE	*p*
Germany, >50 km	0.10	0.17	0.563
Germany, 0–10 km	0.61	0.31	0.050
Germany, 10–20 km	0.18	0.20	0.374
Germany, 20–30 km	0.21	0.19	0.257
Germany, 30–40 km	0.02	0.23	0.925
Germany, 40–50 km	0.12	0.23	0.616
France, 0–10 km	-1.14	0.55	0.039
France, 10–20 km	-1.62	0.54	0.003
France, 20–30 km	-1.03	0.47	0.027
France, 30–40 km	-1.91	0.63	0.002
France, 40–50 km	0.00	0.86	0.998

Notes: Sample: 1–31 May 2022; the placebo period is 18–31 May. Coefficients are relative to French stations more than 50 km from Germany. Station and date fixed effects, holiday controls, and station-clustered standard errors.

Table C.8. Inference for the German diesel premium

Method	SE	*p*	95% interval
Station clusters (2,880)	0.27	<0.001	[-1.49, -0.44]
County clusters (86)	0.54	0.078	[-2.04, 0.11]
County wild cluster bootstrap (86)		0.088	[-2.75, 1.30]
Spatial HAC, 10 km	0.50	0.052	[-1.94, 0.01]
Spatial HAC, 25 km	0.61	0.113	[-2.16, 0.23]
Spatial HAC, 50 km	0.66	0.142	[-2.26, 0.32]
Spatial HAC, 100 km	0.72	0.177	[-2.37, 0.44]

Notes: The estimate is the German-only premium of -0.97 cents per liter (German stations within 10 km of France minus those more than 50 km away; station and date fixed effects, holiday controls; N = 400,250). The wild cluster bootstrap uses Webb weights and 9,999 draws on a regression with date indicators and a station effect. Spatial HAC standard errors use a Bartlett kernel over station distance with the stated cutoff and full serial correlation within stations.

Table C.9. Price-construction sensitivity: tax-equivalent diesel response (%)

Group	Primary	00:00–24:00	3-day cap	7-day cap	14-day cap
Germany, >50 km	79.1	78.0	83.1	80.9	80.5
Germany, 0–10 km	84.9	83.5	89.6	87.2	86.6
France, 0–10 km	4.8	4.9	7.3	5.9	5.2
France, 30–40 km	5.2	5.3	3.5	4.2	4.3

Notes: The primary price averages 06:00–22:00 and has no cap on the age of a carried-forward price; the cap columns drop station-days whose price is older than the stated number of days. Sample sizes are 507,791 (primary), 507,790 (00:00–24:00), 450,130 (3 days), 485,267 (7 days), and 498,864 (14 days).

Table C.10. Diesel panel construction

Construction step	Station-days
German complete 06:00–22:00 station-days	401,328
of which using a carried-forward price	29,502
German days excluded for missing route	1,078
German estimation sample	400,250
German partial-coverage days excluded	60
French estimation sample	107,541

Notes: The German complete-day count precedes routing exclusions. Across Germany and France, the raw files contain 25,484,414 valid price events. France has three ambiguous timestamps (six rows); Germany has none.

CRediT authorship contribution statement

Mats Petter Kahl: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review and editing.

Declaration of interest

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Data availability

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used artificial intelligence tools (Claude (Anthropic), Codex (OpenAI), Grammarly, and Cursor (Anysphere)) to draft and edit the manuscript text, to write, run, and check the analysis code under the author's specifications, and to check numbers and citations against their sources. After using these tools, the author reviewed, verified, and edited all text, code, and results as needed, and takes full responsibility for the content and integrity of the published article.

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